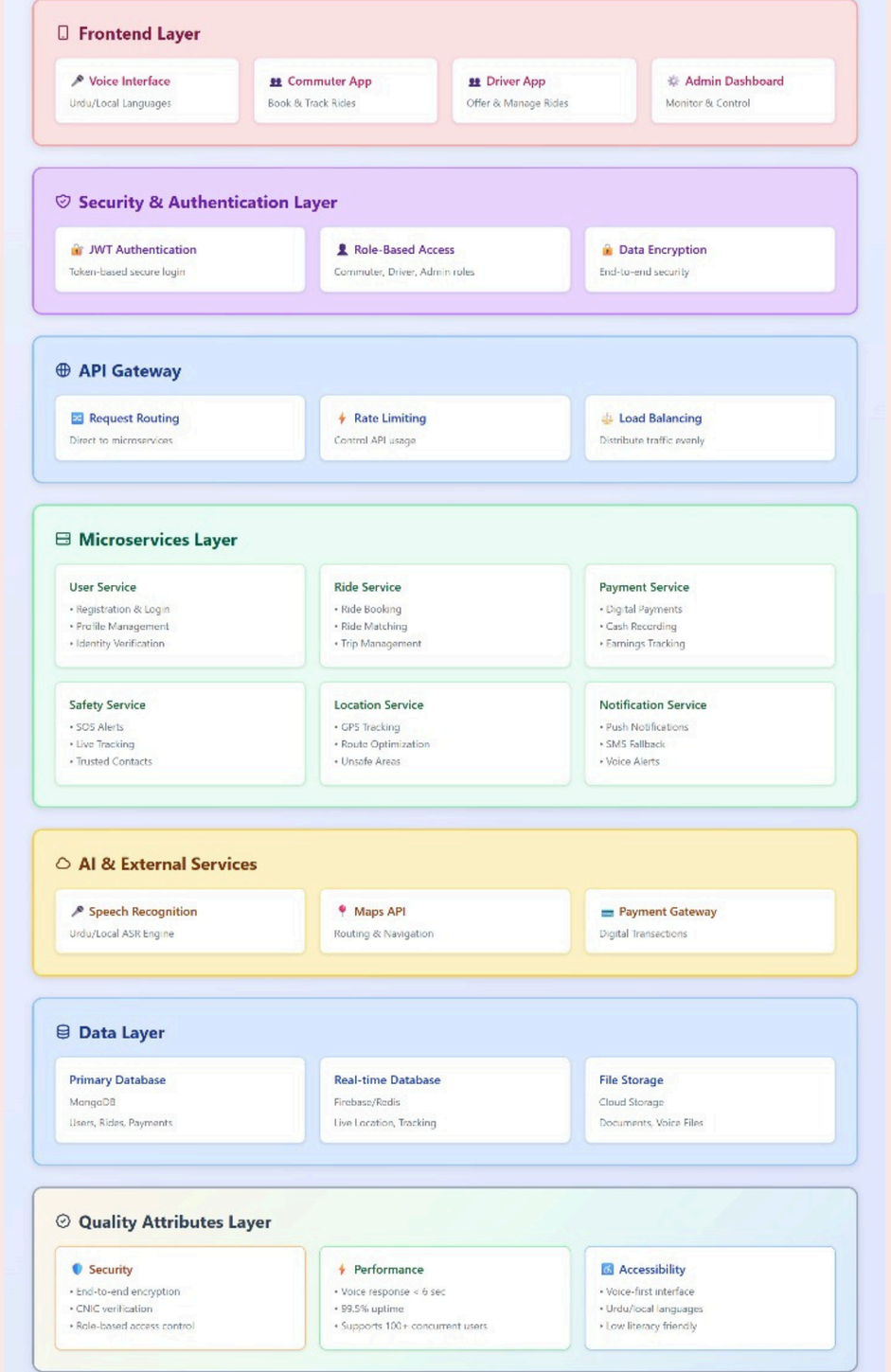
HerSafar

A Voice-Commute App

Group 02

SYSTEM ARCHITECTURE



NON-FUNCTIONAL REQUIREMENTS

Requirement	Architecture Support
1 Achieve ≥**85%** Urdu and ≥**80%** regional dialect accuracy under normal noise conditions	• **AI Layer:** Urdu & dialect-trained ASR engines • **Frontend:** Filters background noise before processing • **Microservices:** Store user language & dialect preferences • **Data Layer:** Feedback logs enable continuous model improvement
2 Application must maintain **99.5%** uptime availability	• **Load Balancing**: Distributes traffic across multiple servers • **Independent Microservices**: Service failures don't crash system • **Rate Limiting**: Prevents system overload during traffic spikes • **Database Redundancy**: MongoDB replication ensures continuous data availability
3 Application must be easy to use for low-literacy users without reading or typing.	• **Frontend:** Voice-first Urdu interface with speech-to-text & text-to-speech • **AI Layer:** Uses Google speech APIs for accurate Urdu voice processing • **Quality Layer:** Simple UI with large icons & minimal text • **Notification Service:** Provides spoken alerts instead of text

Security Requirements as per High-Level Architecture

- **Improper Access Control**

  Role-Based Access Control (RBAC) ensures users access only their permitted functions based on assigned roles (Commuter, Driver, Admin).

- **Insecure Authentication**

  JWT Authentication Service validates credentials and issues secure tokens verified by **API Gateway** on every request.

- **Unauthorized Direct Access to Services**

  API Gateway acts as single entry point with microservices protected in private network, preventing direct unauthorized access.

THANK YOU